

Predictive Model Plan – Student Template

1. Model Logic (Generated with GenAI)

The objective is to predict whether a customer is likely to become delinquent using historical financial and credit data.

Recommended model: Logistic Regression (primary), with Random Forest as an advanced alternative.

Pipeline:

- 1. Load cleaned dataset from Task 1.*
- 2. Select important features (Income, Credit Utilization, Missed Payments, Debt-to-Income Ratio, Credit Score).*
- 3. Preprocess data (handle missing values, encode categorical variables, split into train/test sets).*
- 4. Train Logistic Regression model.*
- 5. Predict probability of delinquency.*
- 6. Classify customers into Low, Medium, or High Risk based on probability.*
- 7. Evaluate model performance using classification metrics.*

Top 5 input features:

- Income*
- Credit Utilization*
- Missed Payments*
- Debt-to-Income Ratio*
- Credit Score*

2. Justification for Model Choice

Logistic Regression is appropriate because predicting delinquency is a binary classification problem.

The model is transparent, easy to interpret, fast to train, and widely accepted in financial services.

It helps explain why a customer is classified as high risk, supporting regulatory compliance and business transparency.

Although more complex models such as Random Forest may improve accuracy, Logistic Regression provides the best balance of explainability, ease of deployment, and reliable performance for Geldium's credit risk assessment.

3. Evaluation Strategy

The model will be evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, and a Confusion Matrix.

- Accuracy measures overall correctness.*
- Precision measures how many predicted high-risk customers are actually high risk.*
- Recall measures how many actual delinquent customers are correctly identified.*
- F1 Score balances Precision and Recall.*
- ROC-AUC evaluates the model's ability to distinguish risky and non-risky customers.*

Fairness and ethics:

- *Compare performance across different customer groups.*
- *Monitor false positive and false negative rates for bias.*
- *Avoid using discriminatory variables.*
- *Retrain and monitor the model regularly to maintain fairness and reliability.*